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$$(1) \det \begin{pmatrix} 4 & 2 & -\lambda & 0 \\ 2 & 1 & 0 & \lambda \end{pmatrix} = \det \begin{pmatrix} 4-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} =$$

$$(4-\lambda)(1-\lambda) - 4 = 0$$

$$\lambda^2 - 5\lambda = 0$$

$$\lambda = 0, 5$$

$$\lambda = 0: A - \lambda I = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} x = 0 \quad x = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{pmatrix}$$

$$\lambda = 5: A - \lambda I = \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} x = 0 \quad x = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix}$$

(2)

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} = \begin{pmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix}$$

P: orthogonal const.

(3)

$\lambda_1 = 0, \lambda_2 = 5$ PSD

1-2

$$B = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix}$$

(1)

$$B'B = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 7 \\ 7 & 9 \end{pmatrix} \quad BB' = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix}$$

(2)

$$\det \begin{pmatrix} 9 & 7 \\ 7 & 9 \end{pmatrix} - \lambda^2 = 0 \Rightarrow \det \begin{pmatrix} 9-\lambda & 7 \\ 7 & 9-\lambda \end{pmatrix} = 0 \Rightarrow \lambda^2 - 18\lambda + 82 = 0 \quad \lambda = 2, 16 \quad \sum \lambda = \text{tr}(B'B)$$

$$\lambda = 16 \Rightarrow (B'B - 16I)x = 0 \Rightarrow \begin{pmatrix} 7 & 7 \\ 7 & 7 \end{pmatrix} x = 0 \Rightarrow \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$V = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

$$\lambda = 2 \Rightarrow (B'B - 2I)x = 0 \Rightarrow \begin{pmatrix} 7 & 7 \\ 7 & 7 \end{pmatrix} x = 0 \Rightarrow \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

$$\sigma_1 = \sqrt{16} = 4 \quad \sigma_2 = \sqrt{2}$$

(3)

$$\det \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix} - \lambda^3 = 0 \Rightarrow \det \begin{pmatrix} 8-\lambda & 0 & 8 \\ 0 & 2-\lambda & 0 \\ 8 & 0 & 8-\lambda \end{pmatrix} = 0 \quad (8-\lambda)(8-\lambda)(2-\lambda) = 0 \quad \lambda = 16, 2, 0 \quad \sum \lambda = \text{tr}(BB')$$

$$\lambda = 16 \Rightarrow (BB' - 16I)x = 0 \Rightarrow \begin{pmatrix} 8 & 0 & 8 \\ 0 & -14 & 0 \\ 8 & 0 & -8 \end{pmatrix} x = 0 \Rightarrow \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$$

$$U = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{pmatrix}$$

$$\lambda = 2 \Rightarrow (BB' - 2I)x = 0 \Rightarrow \begin{pmatrix} 6 & 0 & 8 \\ 0 & 0 & 0 \\ 8 & 0 & 6 \end{pmatrix} x = 0 \Rightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\lambda = 0 \Rightarrow (BB' - 0I)x = 0 \Rightarrow \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix} x = 0 \Rightarrow \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$$

(4)

$$\begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

1-3

(1)

$$\det \begin{pmatrix} 5-\lambda & 3 \\ 3 & 3-\lambda \end{pmatrix} = 0 \rightarrow \lambda^2 - 8\lambda + 16 = 0 \quad \lambda = 8, 2$$

$$\lambda = 2 \quad (2-8I)x=0 \rightarrow \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} x=0 \rightarrow x = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

$$\lambda = 8 \quad (8-8I)x=0 \rightarrow \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} x=0 \rightarrow x = \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}$$

(2)

$$PC1: \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} \quad PC2: \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}$$

(4)

20%

P(그로 크기항에 따라

P(2의 방향의 변산은 설명되지 않는다

따라서 PC2가 설명하는 비율은 20%가
크게 된다

(3)

$$PC1: 8/10 \quad PC2: 2/10$$

2-1

(1)

$$P(A) = 8/16 = 1/2$$

$$P(B) = 4/16 = 1/4$$

$$P(C) = 4/16 = 1/4$$

(2)

$$\frac{1}{2} \cdot \log_2 2$$

$$+ \frac{1}{4} \cdot \log_2 4$$

$$+ \frac{1}{4} \cdot \log_2 4 = \frac{3}{2}$$

(3)

$$\text{Max: } \frac{1}{3} \cdot \log_2 3 \cdot 3 = \log_2 3$$

$$> \frac{3}{2}$$

∴ 현재 데이터가 불확실성

2-2

(2) 두 분포의 엔트로피를 비교

P와 Q가 동일 분포일 때 0이 됨.

(1)

$$D(P||Q) = 0.5 \cdot \log_2 \frac{0.5}{0.25} + 0.3 \cdot \log_2 \frac{0.3}{0.25} + 0.2 \cdot \log_2 \frac{0.2}{0.25} = \frac{1}{2} + 0.3 \cdot \log_2 0.6 + 0.2 \cdot \log_2 0.8$$

≠

$$D(Q||P) = 0.25 \cdot \log_2 \frac{0.25}{0.5} + 0.5 \cdot \log_2 \frac{0.5}{0.3} + 0.25 \cdot \log_2 \frac{0.25}{0.2} = -\frac{1}{4} + 0.5 \cdot \log_2 \frac{5}{3} + \frac{1}{4} \cdot \log_2 1.25$$

$$d(P,Q) \neq d(Q,P) \text{ 옳지 않다}$$

3-1

(1)

$$\bar{x} = 3 \quad S_{xy} = 17$$

$$\bar{y} = 5 \quad S_{xx} = 10$$

(2)

$$\hat{\beta}_1 = \frac{17}{10}$$

$$\hat{y} = 1.7x - 0.1$$

$$\hat{\beta}_0 = 5 - 1.7 \cdot 3 = -0.1$$

(3)

x 1 2 3 4 5

y 2 3 5 6 9

y 1.6 3.3 5 6.7 8.4

e 0.4 -0.3 0 -0.7 0.6

e^2 0.16 0.09 0 0.49 0.36

$$SSB = 1.1$$

$$SSY = 30$$

$$R^2 = 1 - \frac{SSB}{SSY} = \frac{28.9}{30}$$

(4)

제곱합이 $\frac{28.9}{30}$ 을 크기에 상관없이 7.3

크기에 상관없이 7.3

3-2

(1)

$$VIF_{\bar{x}} = \frac{1}{1 - R_{\bar{x}}^2}$$

$$\Rightarrow VIF_1 = \frac{1}{1 - 0.9} = 10$$

$\Rightarrow X_{\bar{x}}$ 를 나머지 X 들로 대체시 R^2

(2)

$VIF = 10$ 은 상당히 다중공선성이 우려되는 수치. 변수 제거 권장.

4-1

(1)

(1,1) (4,4) $\sqrt{8}$

(3,4) " $\sqrt{0.25}$

(5,5) " $\sqrt{2}$

(5,3) " $\sqrt{2}$

(6,4) " $\sqrt{4}$

(3)

2 > $\sqrt{2}$

A B

$\therefore 9 \text{ class} = A$

(2)

(3,5,4) A

(5,5) B

(5,3) B

9 class: B

(4)

한 다중공선성 k 값이 많을수록 k↑ 한 어긋나기 가능성 커짐

weight k를 거리 가중치로 있어 k에 영향

5-1

(1)

	$C_1(2,3)$	$C_2(8,8)$	비교
(2,3)	0	$\sqrt{61}$	C_1
(2,1)	2	$\sqrt{85}$	C_1
(3,2)	$\sqrt{2}$	$\sqrt{61}$	C_1
(8,8)	$\sqrt{61}$	0	C_2
(9,8)	$\sqrt{74}$	1	C_2
(8,9)	$\sqrt{72}$	1	C_2

(2)

$$\text{New } C_1 = \left(\frac{2+2+3}{3}, \frac{3+1+2}{3} \right) = \left(\frac{7}{3}, 2 \right)$$

$$\text{New } C_2 = \left(\frac{8+9+8}{3}, \frac{8+8+9}{3} \right) = \left(\frac{25}{3}, \frac{25}{3} \right)$$

(3)

재배정 결과 동일 : 알고리즘 수렴함

5-2

(1)

P_1 - 4개 (P_1, P_2, P_3, P_4) P_5 - 3개 (P_5, P_6, P_8) P_9 - 1개

P_2 - 4개 (") P_6 - 3개 (P_5, P_8)

P_3 - 4개 (") P_7 - 2개 (P_8)

P_4 - 4개 (") P_8 - 4개 (P_5, P_6, P_7, P_8)

(2)

Core : $P_1, P_2, P_3, P_4, P_5, P_6, P_8$

border : P_7

Noise : P_9

(3)

K-means는 사전 지정된 K개 군집으로 모든

점을 배정하지만, DBSCAN은 밀도 기반으로

밀도 중심 수렴을 군집수가 정해짐

Cluster 1 : { P_1, P_2, P_3, P_4 }

Cluster 2 : { P_5, P_6, P_8, P_7 }